Catch-up on the second milestone

What happened after the booklet froze, 25 and 26 October 2026

Verdict

The booklet is unchanged since you froze it, so read it as it is. This reading covers what happened after it. The second milestone, mappings to known hard problems, ran three spaces. Two closed with no blocking challenge. The third, computational limits, ended with one open challenge on one theorem, an implementation detail of its arithmetic model. The milestone produced 18 accepted results and one sketch. The integrator re-derived 16 of the 18 proofs, and `lab/theory/integration-log.md` lists the other two and what it took on trust. Five results deserve your reading, and six decisions wait for you. Nothing of this milestone is in the booklet yet. It enters after your reading, and nothing since the push of 25 October is committed.

The discussant setup also changed, on your instructions of that evening. sol (gpt-6.1-sol at high reasoning) replaced astra, after a Codex update to version 0.164.0. argon's failures had one cause, read off its transcripts. A whole-round review thought for 22,000 to 27,000 tokens without streaming any, and agy cuts a stream that stays silent for five minutes. argon now writes its review one object at a time, with permissions bypassed and every call audited. Its quota ran out before a round could finish under the new protocol, so every round of this milestone used sol, and the protocol is not yet seen to finish.

1. A same-units gap in a natural procedure

Statistical limits, `thm-same-units-gap-gaussian-mechanism` and `thm-train-access-price-frontier`.

The procedure is the noisy mean estimator of Haghifam, Smith and Ullman, with noise $\rho\Sigma^{1/2}\xi$ calibrated to the population covariance. The trainer spends n samples and one run. With Train access and $\rho = n^{-a}$, $0 < a < \frac{1}{2}$, every membership attack that meets their criterion spends more than $\frac{c_2}{2}(n+n^2\rho^2)$ in samples plus Train calls, and an efficient attack attains it. So the attacker pays the factor $\Theta(1 + n\rho^2)$ over the trainer, which is the headline the adopted objective asks for, in a natural procedure. Train access moves the price between the two resources. Reference samples fall from $\Theta(n^{2-2a})$ to $\Theta(n^{2a})$, paid with $\Theta(n^{2-2a})$ Train calls, and the combined price is unchanged. Status. Proven, under the modelling hypothesis of decision 1 below.

2. Extraction costs twice the trainer's exponent

Hard-problem mappings, `thm-extraction-exponent-price` with `lem-level-set-moment-balance`.

Take labels $Y = Z\varepsilon$ with $E[Y \mid Z] = |Z|e^{-Z^2/2}$ on $Z = \langle u, x\rangle$. Their generative exponent is 2, so the trainer recovers the direction from $O(d)$ records and releases the Bayes predictor $m(\langle\hat{w}, x\rangle)$. The release's own

link has generative exponent 4. The reason is one identity: on each level set of $|z|\phi(z)$, the Jacobian weights $1/(1-a^2)$ and $1/(b^2-1)$ give a Gaussian its first three moments exactly. Extracting the release from noisy answers on random inputs therefore needs VSTAT($c\,d^2$) for statistical-query extractors, and $O(d^2)$ answered pairs suffice. The attacker's exponent is twice the trainer's. No cryptographic primitive enters, and the hard instance is a label law. Status. Proven for the SQ lower bound and the sample upper bound. The comparison of the trainer's records with the extractor's VSTAT parameter is heuristic, and chosen inputs break the separation.

3. A blind server learns records only up to their symmetry

Statistical limits, `thm-blind-orbit-limit`, `thm-blind-orbit-price`, `prop-relu-client-second-moment-identifies-weights`.

An unbounded server without the seed learns the records of a split protocol only up to the symmetry group of the record law, for frozen or trained clients. For Gaussian records it cannot place even one record within $\sqrt{d}/(2\sqrt{2})$ except with probability $e^{-\Omega(d)}$. Up to that rotation, the minimax price is $\Theta(d^2/N)$ per record, so relative accuracy ϵ costs $\Theta(d/\epsilon^2)$ representations, and empirical whitening attains it. For a Gaussian ReLU client, the population second moment identifies the weights up to rotation through the arc-cosine kernel, so the orbit is reached in polynomial time. Status. Proven. The ReLU result is at the population level, and its sample price is open.

4. Blind reconstruction is lattice-hard, in exact arithmetic

Computational limits, `thm-coarse-reconstruction-lattice-hard`, `prop-blind-lll-label-recovery`, `thm-zlip-reductions`.

Reconstructing pancake records to any constant correlation, without the seed, is as hard as homogeneous CLWE. The proof makes two independently rotated runs agree on the hidden direction (`lem-agreement-detection`). This closes the coarse-reconstruction problem left open in the first milestone. With exactly discrete hidden labels, the two attacker games separate on one task. A polynomial-time server reads the labels off $d+1$ representations by LLL, while every statistical-query tester needs $2^{d^{\Omega(c)}}$ queries. For quantized Gaussian records, blind reconstruction is at least as hard as average-case ZLIP, and for a linear client no harder. Status. Proven in an exact-arithmetic model of computation, the ZLIP theorem a sketch. sol accepted its mathematics, and one challenge stays open on how the model encodes real widths and bounds expected running time.

5. What a mapping is, and one corrected literature mapping

Hard-problem mappings, `def-hard-problem-mapping`, `prop-learner-side-sq-transfer`, `lem-sparse-pca-backdoor-width-price`.

A mapping now has a fixed form: procedure, attack task, hard problem, direction, prices on both sides, regime, carrier of the hardness, and experiment. An SQ lower bound on the learner binds every unbounded white-box attacker of its output. This corrects a typographical sign in Feldman et al., Theorem 3.16. The sparse-PCA backdoor of Goldwasser, Kim, Vaikuntanathan and Zamir, Appendix A, either shows up in time $O(md)$ to a coordinate-variance test, or its key buys at most a factor $\sqrt{1+\theta}$ over a

key-free input. Its activation condition also fails at $\theta = 1$, $\lambda = 1.1$, $\tau = 1$. Their main construction, Theorem 6.4 under CLWE, is untouched. Status. Proven.

Decisions for you

1. Train access. Under the Gaussian mechanism calibrated to the population covariance, Train calls leak that covariance, and the leak is what moves the sample price. Is this the right model for an attacker with Train access?
2. Exact arithmetic. The results of section 4 hold in an exact-arithmetic model. Accept them in that model for now, with the transfer to Turing machines as an open problem, or ask for the finite-precision analysis first?
3. Learning-based extraction. The separation of result 2 holds for extraction from random inputs and disappears with chosen inputs. Is that a game you want priced?
4. The carrier rule. The mapping definition counts hardness carried by the record law, by the honest randomness of a standard procedure, and by a trainer's planted choice. It excludes only a key-dependent change the designer makes for security. Is this your reading of "a cryptographic plug-in does not count"?
5. The ZLIP sketch. Spend a round on its open implementation challenge, or leave it as a sketch?
6. The booklet. Merge this milestone into it after your reading, and in what form. The spaces proposed a simplified merge of about fifteen objects.

Where things are

The round files are in `lab/theory/spaces/<space>/round-NN/`, and the full statements and proofs are in the builder files named in each result above. The registers in `sink/formalizing/` now hold 647 entries and 80 searches. The integrator's checks are in `lab/theory/integration-log.md`. The milestone's ledger, `lab/theory/GATES.md`, has three gates open: the booklet merge, which waits for you, the convergence of computational limits, and an argon round under the new protocol.